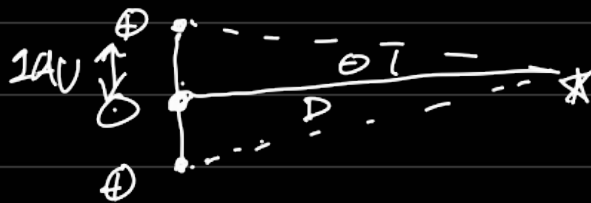


* Distance Modules:



$$s = r\theta, \quad \theta \ll 1$$

$$\boxed{\frac{1AU}{D} = \theta}$$

$$\boxed{m - M = 5 \log(D) - 5}$$

$$\rightarrow \boxed{\frac{I_2}{I_1} = (100)^{\frac{1}{5}(m_2 - m_1)}}$$

$$\log_{10} \left(\frac{I_2}{I_1} \right) = \frac{1}{5} (m_2 - m_1) \log_{10}(100)$$

$$\log_{10}(10^2) = 2$$

$$\log_{10}(10^2) = 2 \log_{10}(10) = 2$$

$$\log \left(\frac{I_2}{I_1} \right) = \frac{2}{5} (m_1 - m_2)$$

$$\frac{5}{2} \log(I_2/I_1) = M_1 - M_2$$

$$\boxed{2.5 \log(I_2/I_1) = M_1 - M_2}$$

$$I_\lambda \propto \frac{1}{D^2}$$

$$\bullet I_\lambda = \frac{L}{4\pi D^2}$$

* Stat Mech:

$$\bullet \text{ Partition function: } Z = \sum e^{-E_n/k_B T}$$

$$\bullet \text{ Prob. of a particle having energy } E_m$$

$$\boxed{P(E_m) = \frac{e^{-E_m/k_B T}}{Z}}$$

$$\bullet \text{ Equipartition theorem:}$$

$$\langle E \rangle = \frac{N}{2} k_B T \quad (\text{Non-rel. particles})$$

$$\langle E \rangle = N k_B T \quad (\text{rel. gas})$$

→ Non-rel particle in ISM:

$$\frac{1}{2} m v_{rms}^2 = \frac{3}{2} k_B T$$

3: Mon. gas has 3. D.O.F.

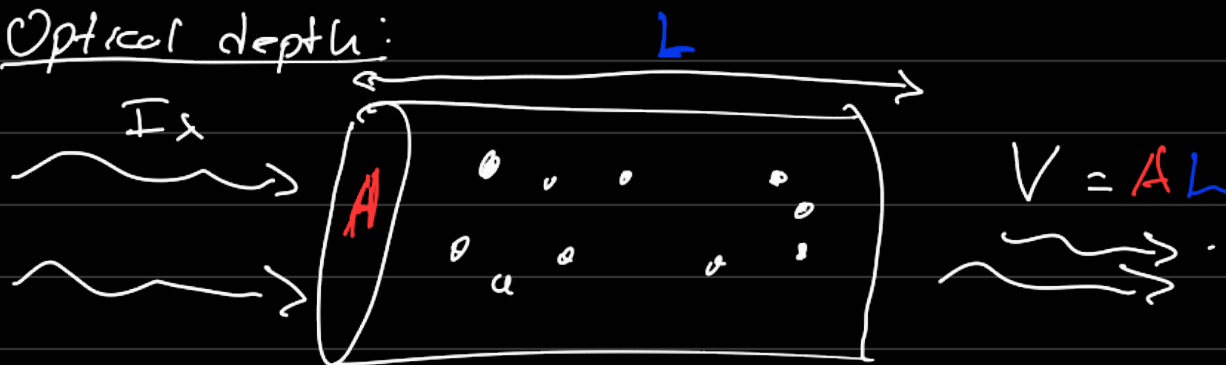
$$v_{rms} = \sqrt{\frac{3k_B T}{m}}$$

* Useful: Use dim analysis

• S_* : -5

$$N_\gamma = S_* \pm$$

Optical depth:



Each part. has a cross-section

$$\sigma = \pi R^2 \text{ (single)}$$

• N - total inside the volume

$$\sigma_{tot} = N\sigma = nV\sigma = nLA\sigma$$

• $n = \frac{N}{V}$

$$(I_{obs}) = \left(\frac{\sigma_{tot}}{A} \right) I_\lambda = \left(\frac{nLA\sigma}{A} \right) I_\lambda$$

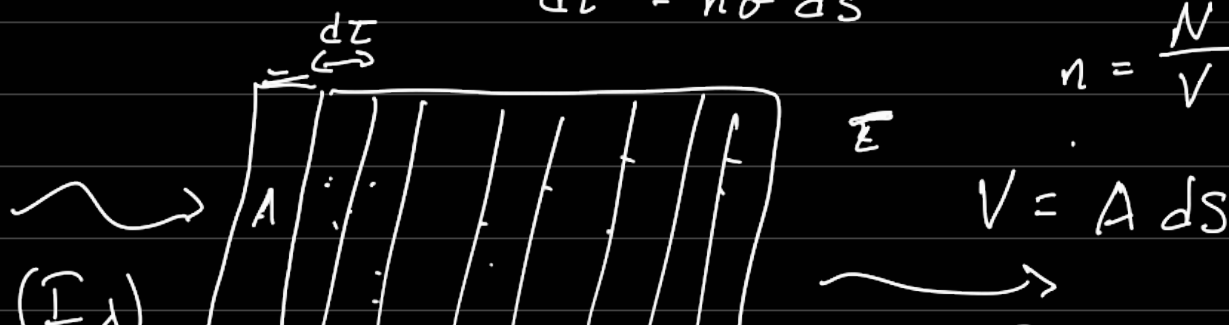
$$= \underline{nL\sigma} I_\lambda$$

* $\tau = nL\sigma$ * (optical depth)

$$L = S$$

$$d\tau = n\sigma ds$$

$$n = \frac{N}{V}$$





$$\sigma_{tot} = N\sigma = nA\sigma ds$$

$$n: \frac{N}{V} \quad m \rightarrow \quad \rho = m n = \frac{(Nm)}{V}$$

$$n = \rho/m$$

$$\sigma_{tot} = \frac{\rho}{m} A \sigma ds = \left(\frac{\sigma}{m} \right) \rho A ds$$

← cross-section

← mass

$$\boxed{K_\lambda \equiv \frac{\sigma}{m}} \text{ (abs. coeff)}$$

$$\sigma_{tot} = K_\lambda \rho A ds$$

$$dI_\lambda = - \left(\frac{\sigma_{tot}}{A} \right) I_\lambda = -K_\lambda \rho I_\lambda ds$$

$$\boxed{dI_\lambda = -K_\lambda \rho I_\lambda ds}$$

$$* \quad dI_\lambda = -K_\lambda \rho I_\lambda ds \quad (\text{separable})$$

$$\int \frac{dI_\lambda}{I_\lambda} = \int -K_\lambda \rho ds$$

$$(\text{RHS}): -K_\lambda \rho s + C$$

$$(\text{LHS}): \ln(I_\lambda) + C'$$

$$\ln(I_\lambda) + C' = -K_\lambda \rho s + C$$

$$C - C' = \tilde{C}$$

$$\ln(I_\lambda) = -K_\lambda \rho s + \tilde{C}$$

$$I_\lambda(s) = \exp(-K_\lambda \rho s + \tilde{C})$$

$$= e^{\tilde{C}} e^{-K_\lambda \rho s}$$

$$\tilde{C} = \ln(I_0)$$

$$e = I_{\lambda}(0)$$

$$s=0$$

$$e^0 = 1$$

$$I_{\lambda}(s) = I_{\lambda}(0) e^{-\kappa_{\lambda} s}$$

$$\begin{cases} d\tau = \rho \kappa_{\lambda} ds \\ I_{\lambda} = I_{\lambda}(0) e^{-\tau} \end{cases}$$

* $\ln\left(\frac{I_{\lambda}}{I_{\lambda}(0)}\right)$

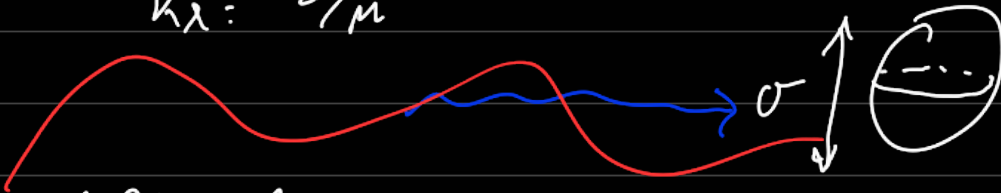
A_{λ} : extinction

$$M_{\text{true}} = M_{\text{obs}} + A_{\lambda}$$

$$* m - M = 5 \log(D) - 5 + A_{\lambda} *$$

• Interstellar reddening:

$$\kappa_{\lambda} = \sigma/n$$



* Blue light get scattered

- $A_{\lambda} \sim 1/\lambda$ (Int. red)

- $A_{\lambda} \sim 1/\lambda^4$ (Rayleigh scattering)

$$* E(\lambda - V) = A_{\lambda} - A_V$$

V : Band
(red)

* $H II$:

$$E_n = -13.6 \frac{1}{n^2}$$

$$n \rightarrow \infty$$

$$\underline{E_{\infty} = -13.6}$$

